

Cloud Architecting – Week 6

Total questions: 20

Worksheet time: 20mins

Name

Class

Date

1. With Amazon Virtual Private Cloud (Amazon VPC), what is the smallest size subnet you can have in a VPC?
a) /26
b) /28
c) /24
d) /30
2. With Amazon Virtual Private Cloud (Amazon VPC), what is the maximum size IP address range you can have in a VPC?
a) /24
b) /30
c) /16
d) /28
3. You need to allow resources in a private subnet to access the internet. Which of the following must be present to enable this access?
a) NAT gateway
b) Security groups
c) Network access control lists
d) Route tables
4. Which AWS networking service enables a company to create a virtual network within AWS?
a) AWS Config
b) Amazon Virtual Private Cloud (Amazon VPC)
c) Amazon Route 53
d) AWS Direct Connect
5. Private subnets have direct access to the internet.
a) True
b) False
6. Which component of AWS global infrastructure does amazon CloudFront use to ensure low-latency delivery?
a) AWS edge locations
b) AWS Regions
c) AWS Availability Zones
d) Amazon Virtual Private Cloud (Amazon VPC)
7. Which of the following is an optional security control that can be applied at the subnet layer of a VPC?
a) Security group
b) Web application firewall
c) Firewall
d) Network ACL

8. What happens when you use Amazon Virtual Private Cloud (Amazon VPC) to create a new VPC?
- a) A main route table is created by default.
 - b) An internet gateway is created by default.
 - c) Three subnets are created by default: one for each Availability Zone.
 - d) Three subnets are created by default: in one Availability Zone.
9. Which of the following can be used to protect Amazon Elastic Compute Cloud (Amazon EC2) instances hosted in AWS?
- a) AMI
 - b) Security group
 - c) internet Gateway
 - d) All of the above
10. You are a solutions architect who works at a large retail company that is migrating its existing infrastructure to AWS. You recommend that they use a custom VPC. When you create a VPC, you assign it to an IPv4 Classless Inter-Domain (CIDR) block of 10.0.1.0/24 (which has 256 total IP addresses). How many IP addresses are available?
- a) 246
 - b) 251
 - c) 250
 - d) 256
11. Which use case indicate that a non-relational database might be a better solution than a relational database? (Select TWO.)
- a) Horizontal scaling for massive data volume
 - b) Data with unpredictable attributes
 - c) Strong read-after-write consistency
 - d) High availability and fault tolerance
12. Which statement that compares a database service tat Amazon AWS manages with a database on an Amazon EC2 instance is true?
- a) You do not need to configure backups for a database on a managed database service.
 - b) AWS manages DB patches for a database on a managed database service.
 - c) AWS manages operating system (OS) patches for a database on an EC2 instance.
 - d) You do not need to configure backups for a database on an EC2 instance.
13. Which examples are good use cases for Amazon Relational Database Service (Amazon RDS)? (Select THREE.)
- a) Thousands of distributed concurrent writes per second
 - b) An application that requires the database to enforce syntax rules
 - c) An application that requires complex joins of data
 - d) Running a Microsoft SQL Server in AWS
14. A small company is deciding which service to use for an enrolment system for their online training website. Choices are MySQL on Amazon EC2, MySQL on Amazon RDS, and Amazon DynamoDB. Which combination of use cases suggests using Amazon RDS? (Select THREE.)
- a) The data is highly structured.
 - b) Student, course, and registration data are stored in many different tables.
 - c) The enrolment system must be highly available.
 - d) The company doesn't want to mange database patches.

15. Which scenarios are good use cases for Amazon DynamoDB? (Select THREE.)
- a) Database for serverless architecture
 - b) Applications that require ACID transactions
 - c) Document database for JavaScript Object Notation (JSON)-based documents
 - d) Binary large object (BLOB) storage
16. A small game company is designing an online game, where thousands of players can create their own in-game objects. The current design uses a MySQL database in Amazon RDS to store data for player-created objects. Which use cases suggest that DynamoDB might be a better solution? (Select TWO.)
- a) Unpredictable attributes for player-created objects
 - b) Large number of player-created objects, each with different attributes
 - c) Quick search and retrieval of player-created objects
 - d) High amount of read activity on player-created objects
17. Which techniques should you use to secure an Amazon Relational Database Service (Amazon RDS) database? (Select THREE.)
- a) Security group to control network access to individual instances
 - b) A virtual private gateway (VGW) to filter traffic from restricted network
 - c) A virtual private cloud (VPC) to provide instance isolation and firewall
 - d) Encryption to protect sensitive data
18. Which technique should you use to secure Amazon DynamoDB? (Select THREE)
- a) An Amazon VPC gateway endpoint to prevent traffic from traversing the internet
 - b) AWS IAM policies to define access at the table, item, or attribute level
 - c) Security groups to control network access to individual instances
 - d) Encryption to protect sensitive data
19. A company wants to migrate their on-premises Oracle database to Amazon Aurora MySQL. Which process describes the high-level steps?
- a) Use AWS Database Migration Service (AWS DMS) to migrate from the Oracle database to Amazon Aurora MySQL
 - b) Use AWS Database Migration Service (AWS DMS) to migrate the data, and then use AWS Schema Conversion Tools to convert the schema.
 - c) Use AWS Schema Conversion Tools to convert the schema, and then use AWS Database Migration Service (AWS DMS) to migrate the data.
 - d) use AWS Schema Conversion Tools to synchronously convert the schema and migrate the data.
20. You must perform a heterogeneous migration from your on-premise facility to a database in a virtual private cloud (VPC). You will use AWS Snowball Edge and AWS Database Migration Service (AWS DMS). At which point do you use AWS Schema Conversion Tool (AWS SCT)?
- a) At the start, to extract the source database into the Snowball Edge, before shipping the device
 - b) After extracting the data from the source database by using AWS DMS, but before shipping the Snowball Edge
 - c) After the data is in the VPC, but before using AWS DMS to load the data into the target database
 - d) After using AWS DMS to load the data into the target database in the VPC

Answer Keys

1. b) /28
2. c) /16
3. a) NAT gateway
4. b) Amazon Virtual Private Cloud (Amazon VPC)
5. b) False
6. a) AWS edge locations
7. d) Network ACL
8. a) A main route table is created by default.
9. b) Security group
10. b) 251
11. a) Horizontal scaling for massive data volume
- b) Data with unpredictable attributes
12. b) AWS manages DB patches for a database on a managed database service.
13. b) An application that requires the database to enforce syntax rules
- c) An application that requires complex joins of data
- d) a highly structured. Microsoft SQL Server in AWS
14. b) The data is highly structured. Student, course, and registration data are stored in many different tables.
- c) This company doesn't want to manage database patches.
15. a) Database for serverless architecture
- b) Applications that require ACID transactions
- c) Documents for JavaScript Object Notation (JSON)-based documents
16. a) Unpredictable attributes for player-created objects
- b) Large number of player-created objects, each with different attributes
17. a) Security group to control network access to individual instances
- b) A virtual private cloud (VPC) to provide instance isolation and firewall
- c) Encrypt sensitive data
- d) Amazon VPC gateway endpoint to prevent traffic from traversing the internet
18. a) An AWS IAM policy to define access at the table, item, or attribute level
- b) Encryption to protect sensitive data
19. c) Use AWS Schema Conversion Tools to convert the schema, and then use AWS Database Migration Service (AWS DMS) to migrate the data.
20. a) At the start, to extract the source database into the Snowball Edge, before shipping the device

